

Relationships between executive functions and vocabulary knowledge in Spanish-speaking children with and without developmental language disorder

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ARTICLE INFO

Keywords:

Developmental language disorder

Executive functions

Vocabulary size

Vocabulary depth

ABSTRACT

Developmental language disorder (DLD) is often associated with deficits in executive functions (EFs). One common area of language difficulty in DLD is the development of vocabulary knowledge and it has been suggested that EF abilities may be linked to this difficulty. However, an explanation for this relationship remains unclear. The rationale for the current study examined the relationship between EFs and two aspects of vocabulary knowledge, receptive vocabulary size and vocabulary depth, in Spanish-speaking children with and without DLD. Vocabulary skills, the EFs of interference control, response inhibition, verbal and nonverbal working memory, and switching, were assessed in a total of 204 children aged 6–8 years: 105 with DLD and 99 with typical development (TD). Relationships were assessed using multiple regression models and path analysis, including EF as predictors, receptive vocabulary size and vocabulary depth as outcome or mediating variables, and age and non-verbal intelligence as covariates. In children with DLD, the EFs verbal working memory and switching were directly associated with receptive vocabulary size and indirectly with vocabulary depth. This indirect relationship was mediated by receptive vocabulary size. In contrast, no EFs were associated with receptive vocabulary size, and vocabulary depth in the TD group. These results suggest that verbal working memory and cognitive flexibility play a role in the cognitive mechanisms linked to vocabulary development in school children with DLD. Intervention programs aimed at improving vocabulary in this population should include EF activities.

1. Introduction

Developmental language disorder (DLD) is a disruption to the acquisition and use of language that is not explained by other conditions, such as neurological problems, auditory deficit, motor impairment, severe environmental deprivation, or intellectual disability (Leonard, 2014). DLD is reported in approximately 7% of the child population and across many languages (Tomblin et al., 1997). In general, the diagnosis of DLD is confirmed when language difficulties persist beyond the age of five and negatively impact

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school development (American Psychiatric Association, 2013; Bishop et al., 2017).

DLD affects the development of different language domains, mainly vocabulary knowledge, grammar, and discourse (American Psychiatric Association, 2013; Bishop et al., 2017). In the DLD literature there is much interest in delays in acquiring vocabulary knowledge given the importance of this language domain for the development of other linguistic skills (McMurray et al., 2019), as well as the role it plays in academic learning more generally (Green & McLachlan, 2024). There is considerable evidence that children with DLD have delays in two aspects of vocabulary knowledge: receptive vocabulary size, i.e. the number of words a person understands (Li & Kirby, 2015), and vocabulary depth i.e. how much one knows about a word (Li & Kirby, 2015). Regarding receptive vocabulary size, several studies have found that children with DLD understand fewer words than typically developing (TD) children (e.g., Rice & Hoffman, 2015). In relation to vocabulary depth, children with DLD provide definitions with less detailed information than their TD peers (e.g., Marinellie & Johnson, 2002; McGregor et al., 2013). Receptive vocabulary size is expected to be related to vocabulary depth in children with TD and DLD. While there is some debate on the direction of the relationship between these two aspects of vocabulary (e.g., Li & Kirby, 2015), there is strong evidence that early receptive vocabulary size predicts later ability to provide definitions (vocabulary depth) (e.g., Nippold, 2007; Stahl & Nagy, 2006).

Currently there is some discussion that children with DLD have wider areas of cognitive differences other than just language (Bishop et al., 2017). This is seen especially in cognitive processes related to the control and processing of novel information i.e. executive functions (EFs) (Kapa et al., 2017). If the EF system and DLD are related, this suggests that delays in language, including vocabulary development, could be related to difficulties with EFs. This association follows from the more general hypothesis that EF and vocabulary support each other during typical language development (Slot & Von Suchodoletz, 2018). While the nature of the relationship (EF leads vocabulary, vice versa or a third latent variable affecting both abilities) is still a matter of debate (see Blom & Boerma, 2019; Schmitt et al., 2019), a significant number of studies support the idea that EFs predict vocabulary development in both TD (e.g., Ebert et al., 2013; Weiland et al., 2014) and DLD groups (Blom & Boerma, 2019; Everaert et al., 2023; Vugs et al., 2017).

There is much debate about the organization of the different abilities termed EFs (e.g., Friedman & Miyake, 2017). However, there is agreement that EFs include a range of related skills, which, from the onset of school age, begin to organize themselves into distinct constructs (e.g., Lehto et al., 2003). Miyake et al. (2000) proposed a model with three independent but interrelated EF components: inhibitory control, working memory, and cognitive flexibility. Inhibitory control is the ability to deliberately constrain automatic, dominant, or prepotent responses in achieving a goal. Working memory is used to maintain, manipulate, and temporarily process information while performing complex tasks (Baddeley, 2010). Finally, cognitive flexibility is the ability to change quickly from one response to another using alternative strategies (Anderson, 2002).

The rationale for this study is based on the notion that EFs support the acquisition of vocabulary knowledge. In particular, previous research highlights the role of inhibitory control and working memory in vocabulary development. According to these proposals, inhibitory control plays a crucial role in helping typically developing children focus on relevant linguistic information while filtering out distractions—such as irrelevant sounds or background noise—when learning new words (Weiland et al., 2014). In addition, when children are acquiring new words, inhibitory control supports the filtering of semantic and phonological competitors within the already acquired lexicon that if uncontrolled may interfere with new word learning. (Gandolfi & Viterbori, 2020; McClelland & Elman, 1986). Therefore, for children with poor inhibitory control (an EF difficulty), a reduction in the ability to suppress irrelevant stimuli could lead to delays and misidentification of words. For instance, in tasks requiring word learning in the presence of distractors, children with lower inhibitory control might be prone to errors, mistaking one word for another or recalling incorrect associations.

Similarly, much research has argued that working memory is involved in vocabulary development and helps the child to match words they are listening to with visual information in the environment they are scanning (Gathercole, 2006). Working memory also supports children's processing of incoming verbal signals by comparing them in real-time with those stored in long-term memory (Archibald, 2017). A delay in working memory development would make the task of matching newly heard vocabulary to referents much more difficult. Reduced working memory capacities would mean the new word-form would fade from the phonological loop before the child could pick out a possible referent in the environment (Gathercole, 2006). Poor working memory would mean making decisions in real-time, if the new word is already part of the lexicon in long-term memory, more cognitively taxing (Archibald, 2017). Although research on the role of cognitive flexibility in vocabulary development is limited, existing evidence suggests that this executive function facilitates word learning, particularly in contexts that require switching between multiple tasks simultaneously (Hill & Wagovich, 2020). From this point of view, a difficulty in cognitive flexibility would impact vocabulary knowledge and so we included this measure in our study.

This study also builds on the idea that as children with typical development mature and their lexicons expand, the processing of language becomes more automatic and less dependent on these EFs (e.g., Hartsuiker & Moors, 2017). In contrast, it has been suggested that children with DLD during the same time period continue to be dependent on the EFs as a compensatory mechanism when processing vocabulary (e.g., Blom & Boerma, 2019; Kalliontzis et al., 2022). The current study will test this possibility. In addition, this study is based on the idea that vocabulary knowledge is not a unified construct (Koizumi & In'nami, 2020) but encompasses both size and depth and interactions of both characteristics (Koizumi & In'nami, 2020; Nippold, 2007; Stahl & Nagy, 2006).

Although previous studies support the idea that EFs contribute to vocabulary processing in children with and without DLD, the specific role of each EF in the two aspects of vocabulary—receptive vocabulary size and vocabulary depth—remains unclear. The present study aims to measure the relationship between several EF skills and the two aspects of vocabulary, size and depth, in children with typical and delayed language. This endeavor may contribute to a broader understanding of how specific EF deficits may underlie language learning challenges in DLD, with potential implications for targeted interventions.

1.1. Executive functions (EFs) in DLD children

Previous studies have shown that children with DLD have lower EF performance in inhibitory control, working memory, and cognitive flexibility compared with TD children (e.g., Pauls & Archibald, 2016; Vugs et al., 2014). However, it is also the case that children with DLD perform differently across subcomponents of each EF (e.g., Blom & Boerma, 2020; Lukács et al., 2016). From the initial proposal by Miyake et al. (2000), different subcomponents associated with inhibitory control (interference control and response inhibition), working memory (verbal working memory: VWM, and nonverbal working memory: NVWM), and cognitive flexibility (switching and fluency) have been identified (Diamond, 2013; Friedman & Miyake, 2004). Several authors have found that fluency tasks are related to other cognitive abilities, e.g., language, long-term memory, and processing speed (e.g., Whiteside et al., 2016). For this reason, verbal fluency was excluded as an EF subcomponent in our research.

In terms of inhibitory control, children with DLD have difficulties with the more specific measures of interference control (i.e. the ability to resist a distracting stimuli: Ebert & Kohnert, 2011), and response inhibition (i.e. the suppression of actions that are inappropriate in a given context: Kapa et al., 2017). Regarding working memory, there is a broad consensus that verbal working memory (VWM) is particularly poor in DLD (Archibald & Gathercole, 2007; Vugs et al., 2014). The nonverbal aspect of working memory (NVWM) has been less studied in the DLD population, and there is less consensus concerning results (Lum et al., 2012; Vugs et al., 2014). Finally, regarding cognitive flexibility, most previous studies report no differences between the DLD and TD groups in switching tasks (e.g. Henry et al., 2012; Im-Bolter et al., 2006). Pauls and Archibald (2016) conducted a meta-analysis, finding small effect sizes for differences between the TD and DLD groups on this EF.

The differing levels of difficulty observed across EF subcomponents in children with DLD suggested that the contribution of EFs to vocabulary processing may vary depending on the specific subcomponent. Empirical evidence supporting this hypothesis is reviewed below.

1.2. Relationships between EF and vocabulary knowledge in TD children

Most research in this area has been conducted with preschool children and considered only size as a measure of vocabulary knowledge. Also, most studies in the TD population have used general and undifferentiated measures of EF. For example, in preschool TD children, both Everart et al. (2023), and White et al. (2017) demonstrated significant associations between global measure of EF (combined components measured as a latent variable) and receptive vocabulary size. There is a suggestion that EF is less important for vocabulary learning in older school-aged TD children who now have established lexicons, however, there are fewer studies with children in this age group to confirm this. Kaushanskaya et al. (2017) assessed the relationship between the three components of EF (inhibitory control, working memory, and cognitive flexibility) and receptive vocabulary size in school-aged TD children without finding a significant association. These findings align with Blom and Boerma (2019), who propose that the relationship between EF (measured as a single overall measure) and receptive vocabulary size is stronger at younger ages in the TD population. In terms of the relationship between EF and vocabulary depth in TD children, very few studies have been carried out. Dosi and Gavrilidou (2020) and Dosi (2021) investigated the relationship between EFs and word definitions in school-aged children, finding a significant association mainly between VWM and vocabulary depth.

1.3. Relationships between EF and vocabulary knowledge in DLD children

Looking at the relationship between EF and vocabulary knowledge in DLD groups, as in their TD peers, most studies have been conducted with preschool children and consider only the size factor. In this population, probably given their heterogeneous performance across the EF subcomponents, separate measures are often used for EFs. However, almost all studies have focused uniquely on the relationship between VWM and vocabulary knowledge. In these studies, many authors show that VWM is significantly and positively associated (with moderate to large effect sizes) with receptive vocabulary size in preschool DLD children (Horohov & Oetting, 2004; Vugs et al., 2016). The relationship between nonverbal WM (NVWM) and receptive vocabulary size has been less studied in children with DLD. Research by Vugs et al. (2016) found that NVWM and receptive vocabulary size were unrelated in preschool children with DLD. Regarding cognitive flexibility, Kalliontzi et al. (2022) found that a switching task was not related with receptive vocabulary size in preschool children with DLD.

In contrast to their TD peers Blom and Boerma (2019) proposed that the relationship between EF and receptive vocabulary size continues to exist in older children with DLD. That is children with language difficulties carry on being reliant on EF to learn vocabulary beyond the early years. In support of this, Vugs et al. (2017) based on a longitudinal study found that VWM measured at 4–5 years significantly predicted language (including receptive vocabulary size) three years later in children with DLD (aged 7–8 years). Similarly, Jackson et al. (2016) found VWM significantly predicted word recognition in 7 year old school children with DLD. Finally, in children with DLD, as in their TD peers, very few studies have explored the relationship between EF and vocabulary depth. Dosi and Gavrilidou (2020) and Dosi (2021) found that response inhibition and VWM were unrelated to vocabulary depth (measured through the information provided in by word definitions) in school-aged DLD children.

In summary, the majority of studies on TD and DLD focus on receptive vocabulary size rather than vocabulary depth and in samples of younger preschool rather than older school-aged children. Previous studies on the relationship between EF and receptive vocabulary size find a relationship in preschool TD children which may continue to exist in older school-aged children with DLD. In contrast, the relationship between EF and vocabulary depth has only been studied in school-aged children and appears stronger in TD than in DLD children. All previous studies assessed either receptive vocabulary size or vocabulary depth. To date, no studies have evaluated the

relationship between EF with both aspects of vocabulary knowledge in the same group of children. This is despite the known links between receptive vocabulary size and vocabulary depth. Studies also tend to focus only on VWM, even though the different EFs can covary (Miyake et al., 2000). It is important to understand more about how other EF subcomponents play a role in vocabulary development. Alternatively, some studies have grouped several EFs together into a single latent variable, making it difficult to identify direct links between each EF subcomponent and vocabulary knowledge. Finally, most of the previous research on the relationship between EFs and vocabulary knowledge have been conducted in English speaking children, making it difficult to generalize findings to DLD children from another context.

Taking all the previous points into account, the present research aims to investigate the relationship between several EF subcomponents (interference control, response inhibition, verbal and nonverbal working memory, and switching) and two different vocabulary aspects, vocabulary size and depth, in a cross-sectional sample of school-aged Spanish-speaking children with TD and DLD, controlling for age and non-verbal intelligence. As some previous research has suggested that the relationships between EFs and language might differ for children with TD and DLD (Blom & Boerma, 2019; Everaert et al., 2023; Dosi & Gavrilidou, 2020), the present study analyses the relationship in both TD and DLD groups.

The research questions addressed are:

1. Is there a direct relationship between EF subcomponents and receptive vocabulary size in children with TD and DLD, after accounting for participants' age and non-verbal intelligence? If so, which EF subcomponents best explain the variability of receptive vocabulary in each group, TD and DLD?

—**Hypothesis.** Based on previous research in school-aged TD and DLD children (Jackson et al., 2016; Kaushanskaya et al., 2017; Vugs et al., 2017), we only expect an association between EF subcomponents and receptive vocabulary size in the DLD group. We expect VWM to best explain variability of receptive vocabulary size in the DLD group.

2. Is there a direct relationship between EF subcomponents and vocabulary depth in children with TD and DLD, after accounting for participants' age and non-verbal intelligence? If so, which EF subcomponents best explain the variability of vocabulary depth in each group, TD and DLD?

—**Hypothesis.** Based on previous research in school-aged TD and DLD children (Dosi & Gavrilidou, 2020; Dosi, 2021), we only expect an association between EF subcomponents and vocabulary depth in the TD group. In addition, we expect VWM to best explain variability of the vocabulary depth in the TD group.

3. If there is a relationship between EF, vocabulary size and depth in children with TD and DLD, does vocabulary size play a mediating role in the relationship between EF and vocabulary depth?

—**Hypothesis.** Considering that a relationship between EF and receptive vocabulary size is expected only in children with DLD (see first hypothesis) and also that there is a relationship between vocabulary size and depth in this group (e.g., Nippold, 2007; Stahl & Nagy, 2006), we expect that vocabulary size will play a mediating role between EF and vocabulary depth in the clinical language group.

2. Material and methods

2.1. Participants

All children who participated in the study (both DLD and TD groups) came from public schools in Santiago de Chile which followed the same academic curriculum. Students attending these schools come from predominantly low Socioeconomic Status (SES) families. This classification is determined by Chile's Education Quality Agency based on parents' monthly per capita income and educational level. For each child with DLD, a TD peer was selected from the same classroom and no more than five months apart in age.

Children with DLD had a diagnosis from a speech-language pathologist (SLP) from each school using standardized tests, following the guidelines formulated by the Chilean Ministry of Education (Ministerio de Educación de Chile, 2010). These guidelines are, in turn, based on the criteria defined by the International Statistical Classification of Diseases and Related Health Problems (World Health Organization (WHO), 2019) and by the Diagnostic and Statistical Manual of Mental Disorders - V (American Psychiatric Association (APA), 2013). Children between the ages of 6 - 6;11 were assessed for language with the *Test para evaluar procesos de simplificación fonológica* (Test to evaluate phonological processes; Pavez et al., 2008; Cronbach's $\alpha = 0.90$), the *Test Exploratorio de Gramática Española* (Screening Test of Spanish Grammar; Pavez, 2010), which evaluates syntactic development, using a receptive subtest (Cronbach's $\alpha = 0.83$) and an expressive subtest (Cronbach's $\alpha = 0.77$), and the *Test de Comprensión Auditiva de Lenguaje* (Test for Auditory Comprehension of Language; Pavez, 2008; Cronbach's $\alpha = 0.77$), which assesses auditory comprehension of vocabulary, morphology and syntax. From age 7 years, children's language was assessed with the battery *Instrumento de Diagnóstico para los Trastornos Específicos del Lenguaje en la edad escolar* (Diagnostic tool for specific language impairment at school age, Pérez et al., 2015). This test assesses comprehension and expressive language in phonology, syntax, and semantics. It also covers pragmatic skills. All tasks that comprise this battery have a Cronbach's alpha above 0.70. As a complement to the SLP assessment, the children received a qualitative

pedagogical and medical assessment. The diagnosis of DLD is assigned when three minimum requirements are met: a) Assessment by standardized language testing indicates grammatical difficulties (below -2 standard deviation according to syntax test standards), b) the pedagogical assessment indicates learning difficulties associated with DLD, and c) the medical assessment indicates the absence of other disorders that may affect language development. In both age ranges, the total assessment confirm or rule out DLD but do not indicate the degree of severity.

Additionally, the criteria for inclusion for the DLD group were: (1) no previously diagnosed autistic spectrum disorder, intellectual disability, neurological difficulties, or other conditions that could affect language development, based on educational records and teachers reports. (2) normal hearing, (3) typical or corrected vision, (4) non-verbal intelligence within a normal range, assessed with the Raven's Coloured Progressive Matrices Test (Raven, 2005). Two children were excluded for abnormal hearing and one for low performance on Raven's test. Children in the TD group had: (1) Normal language development and academic performance at or above the grade average, reported by their class teacher. In addition, the same inclusion criteria were used as for children with DLD. Although both groups were within the normal range on non-verbal intelligence there was a statistically significant difference on IQ scores derived from the Raven's progressive matrix test ($t(201) = 3.53, p < .001, d = 0.49$) (see Table 1).

A complementary evaluation was conducted by our research team to confirm DLD and TD, using the Core Language Score (CLS) of the Clinical Evaluation of Language Fundamental – Fifth Edition – Spain version (CELF-5; Wiig et al., 2018; standardized version for native Spanish-speaking populations). The Core Language Score (CLS) is a composite score derived from four CELF-5 tasks: Sentence Comprehension, Word Structure, Formulated Sentences, and Recalling Sentences. These four tasks have a reliability coefficient above 0.88. CLS represents general language competence and can determine the presence or absence of DLD with 100% sensitivity and specificity according to the instrument norms (considering a cut-off point of 85 CLS or -1 standard deviation). Based on standardized test scores, DLD was confirmed if children's performance was at or below 85 points on the CLS (1 or more standard deviations below the mean). Seven children were excluded because the diagnosis of DLD given by the school did not match the CELF-5 scores. Conversely, TD was confirmed if performance was above 85 points on the CLS. Four children were excluded because the CELF-5 did not confirm their TD status. We used the CELF-5 Spain standards due to the lack of Chilean norms and the absence of tests other than those already used in schools for the identification of DLD. When comparing both TD and DLD groups on the CLS, significant differences were observed, $t(193)=19.61, p < .001, d = 2.73$ (see Table 1).

To estimate the sample size needed to achieve the desired statistical power (0.80) in our analyses, an *a priori* calculation was performed using R software (R Core Team, 2015, *pwr* package). We estimated a sample size of $n = 95$ for a maximum of seven possible predictors (five EF subcomponents plus two covariates), assuming a medium effect size ($F^2=0.15$), a significance level of $\alpha=0.05$, and a power of 0.80. This sample size is comparable to those in similar studies with DLD (e.g., Everaert et al., 2023; Kalliontzi et al., 2022; Kapa & Erikson, 2020).

A total of 204 children were included and divided into: a) the DLD group, consisting of 105 children (65.7 % male), aged between 74 and 103 months ($M = 85.92, SD = 6.98$), and b) the group of 99 children with TD (47.5 % male), aged 74–104 months ($M = 87.16, SD = 7.36$). When performing an independent *t*-test, there are no differences between the ages of the groups, $t(200)=1.23, p = .220, d = 0.17$ (see Table 1).

This study was approved by the ethics committee of the Pontificia Universidad Católica de Chile (ID 221,207,005) following the ethical requirements proposed by the Helsinki Convention. Parents and children included in the study agreed to participate by informed consent or assent, respectively.

2.2. Procedure

The data were collected from the children at their school in a quiet room with no background noise and normal lighting. Initially, schools provided a list of children who had been diagnosed with DLD and a list of TD children, as observed in the classroom. By reviewing school records and interviewing teachers, the inclusion criteria were confirmed. Then, in the first testing session, an audiologist applied the audiometry screening. In the second session, tests were administered to confirm the diagnosis of DLD or TD (CELF-5 and non-verbal intelligence test). In the third session, EF tests were administered. The research used EF tasks with minimal verbal demands appropriate for children with DLD (Kapa et al., 2017). Finally, in the fourth session, vocabulary tests were administered.

Table 1

Mean and standard deviation of age, non-verbal intelligence, and Core language (CELF-5) for TD and DLD children. Independent samples *t*-tests for group differences are also presented.

Variables	TD ($n = 99$)		DLD ($n = 105$)		T-test		
	M	SD	M	SD	t	p-value	Cohen's D
Age (months)	87.16	7.36	85.92	6.98	1.23	.220	0.17
Non-verbal intelligence ^a	112.14	11.66	106.39	11.59	3.53	< 0.001*	0.49
Core language CELF-5 ^b	94.38	7.19	71.31	9.51	19.61	< 0.001*	2.73

Note. M: Mean; SD: Standard deviation; TD: Typical development; DLD: Developmental language disorder.

^a IQ scores derived from the Raven's progressive matrix test.

^b Standard score on the Core language measure of Clinical Evaluation of Language Fundamental – Fifth Edition – Spanish version (CELF-5).

* Statistically significant differences.

2.3. Measures

2.3.1. Receptive vocabulary size

The Peabody Picture Vocabulary Test-III, Spanish version (PPVT-III) (Dunn et al., 2006), assesses receptive vocabulary size. This version of the instrument has been widely used in recent studies to assess receptive vocabulary development in Spanish-speaking children (including Chilean children), showing excellent validity and reliability indicators (e.g., Ahufinger et al., 2022; Olabarrieta-Landa et al., 2017). The PPVT-III has four pictures per item, accompanied by a spoken target word. The examiner says a word, and the child has to identify the correct illustration representing its meaning. Correct responses receive one point, and incorrect ones receive zero points. The PPVT-III consists of 16 levels, with 12 trials containing four pictures each. The test was completed when the participant made eight or more errors at the same level. The maximum score is 192. This test provides norm data from the age of 2 years six months, with a Cronbach's alpha of 0.91.

2.3.2. Vocabulary depth

The Vocabulary subtest of the Wechsler Intelligence Scale for Children-Fifth Edition (WISC-V) (Rosas et al., 2022b, Chilean version) was conducted to assess vocabulary depth. This subtest consists of 29 items in total. In the first four items, the examiner presents a picture and asks for the meaning. Then, in the other 25 items, the examiner says a word and asks for its meaning. Children are required to give a verbal response on all items. The items are nouns, verbs, and adjectives. The first four items are scored with 1 point if the child's response shows a good understanding of the word and 0 points if the response is incorrect. The other 25 items are scored with 2 points if the child's answer shows a good understanding of the word, 1 point if the answer is correct but fails to incorporate relevant elements, and 0 points if the answer shows an incorrect understanding of the word. The maximum possible score is 54. This subtest has a Cronbach's alpha of 0.89.

2.3.3. EFs subcomponents

An ongoing issue discussed in current research is the impurity that characterizes EF tasks, i.e., a single task may involve more than one EF skill (Friedman & Miyake, 2017; Souissi et al., 2022). Much of the current research on DLD, which uses a differentiated model of EF, recognizes that measures overlap to some extent (e.g., Bloom & Boerma, 2020). Although no test of EF is completely pure therefore, one area of EF is emphasized over others, e.g., tests of verbal working memory, although primarily focused on working memory, involve some overlap with other EFs, e.g., inhibition (Friedman & Miyake, 2017). In the current study we have taken some precautions to diminish the potential impact of overlap. First, we used representative tasks with widely accepted paradigms to assess each EF subcomponent (see Diamond, 2013). In addition, we used tests with high-reliability indicators and because of our DLD sample, measures with low verbal demands. We also ensured that the instruments we administered were adapted for the same context as our sample, with Spanish-speaking populations similar in socioeconomic status and age.

The rationale for the choice of different EF subcomponents was based on the age of the sample (6–8 years). During this period, EFs begin to be organized into separable constructs. Furthermore, previous research reports that children with DLD have variable performances across different EF subcomponents. While VWM and vocabulary knowledge has been studied, the role of the other EF subcomponents is less well understood. This motivated us to measure several EF subcomponents in isolation.

Tasks were taken from the Yellow-Red Battery (Rosas et al., 2022a) used to assess interference control, response inhibition, NVWM, and switching. The Yellow-Red Battery is a standardized instrument with norms between 6 and 11 years of age. This battery uses play activities on an electronic device (Tablet) lasting approximately 45 min (Rosas et al., 2022a). The selection of the task format was based on current technology and game-based formats. Previous research has described this format as leading to more engagement and motivation by this age group of children (e.g., Day et al., 2019). This was confirmed in the current study. Moreover, the Recall of Digits Backward subtest of the Wechsler Intelligence Scale for Children-Fifth Edition (WISC-V) (Rosas et al., 2022b, Chilean version) was conducted to assess VWM. The WISC-V is the most widely used intelligence assessment for children aged 6–16 worldwide. All EF tasks were selected because they have been standardized in Chile, show high rates of validity and reliability. In addition, they require minimal linguistic processing, as they have much visual support in the instructions and during the practice items (except for the VWM test). The structure of the EF tasks ensured that children followed the test instructions before continuing. Only one child with DLD was excluded from the study because they did not understand the instructions in one of the EF tasks. Each EF task is described in the following sections.

2.3.3.1. Interference control. The Arrows task is a version of the Flanker task (Eriksen & Eriksen, 1974). In the center of the screen, as a model, a pink, light blue, or yellow arrow appears, pointing in one of four directions: up, down, left, or right. Then, below the model arrow, three other arrows of different colors appear, pointing in different directions, one with the same direction as the model arrow and the other two with different directions (distractors). There is no relationship between the direction of the arrow and the color. Participants are instructed that each time the model arrow points to the left, right, or upwards, they should touch the arrow pointing in the same direction, even if the color does not match. Each time the model arrow points down, they are told to not touch any of the three arrows. The model arrows appear randomly, one by one, in different directions. This subtest has 36 items; each correct answer is worth one point, so the maximum score is 36 points. In terms of the robustness of the measure, normative data indicates the task is highly correlated with the Flankers test ($r = 0.72$) and discriminates age well (Rosas et al., 2022a). It has a Cronbach's alpha of 0.88.

2.3.3.2. Response inhibition. The Flies task is a version of the stop-signal task (Verbruggen & Logan, 2008). This task also incorporates

the delay of gratification paradigm (Diamond, 2013). On the screen, several flies are shown flying in different directions, and the children have to swat as many of them as possible. The flies make a buzzing sound as they fly, and when swatted, they make a sound that the children find amusing and rewarding. When a green light comes on, the children should continue swatting the flies; however, when the light changes to red, the children should stop swatting the flies. When the traffic light changes color, the participant has to follow the initial rule. The test lasts 2 min and is divided into eight different time slots in which the red or green light appears. The score is calculated by adding the flies swatted in the green interval minus those swatted in the red-light periods. This task has an $r = 0.82$ as a bipartition indicator for reliability.

2.3.3.3. Verbal working memory (VWM). The Recall of Digits Backward subtest presents a series of digits that the participant must repeat in reverse order. The number of digits per set increases progressively. The first series has two digits, and the last has eight digits. This subtest has 18 items; each correct item is scored with 1 point. The maximum score for recall of digits backward subtest is 18 points. This task has a Cronbach's alpha of 0.91.

2.3.3.4. Nonverbal working memory (NVWM). The Visual Farm task is presented on a tablet screen, where a 3×3 grid appears. A sequence of squares in the grid are illuminated and the participant has to touch the squares in the reverse order to that observed. The sequences range from 2 to 10 visual stimuli. This subtest has 18 items; each correct item is scored with 1 point. The maximum score for each working memory subtest is 18 points. This task has a Cronbach's alpha of 0.82.

2.3.3.5. Switching. The Triads task is an adaptation of the Wisconsin Card Sorting Test (Heaton, 1981). Four stimuli appear on the screen, and children have to select three stimuli with the same sorting criteria. These criteria are not made explicit. The possible sorting criteria are size, color, and shape. The tablet used to run the test provides explicit information about correct and incorrect answers depending on the child's answer. After five consecutive correct answers, the classification criterion changes, unbeknownst to the participant, and the participant has to be flexible and find a new classification criterion. The test consists of 21 items, five corresponding to the implicit criterion of color, five to shape, five to size, and the remaining six to a mixed criterion. Participants have three chances to find the category (classification criterion); if they fail, they move on to the next category, but in the analysis these omitted items are included. Failure on each attempt is considered an error. Three failed attempts are considered perseverative errors. One point is awarded for each correct answer on the first attempt, 0.6 on the second attempt, and 0.3 on the third attempt. One point is deducted for each attentional error and 2 points for each perseverative error. Children fail after three incorrect attempts. The maximum possible score for this test was 21 points. The robustness of the measure is high. Normative data show that this task is correlated with the Modified Card Sort Test (a child version of the Wisconsin Card Sorting Test; Nelson, 1976) ($r = 0.42$) and discriminates age well (Rosas et al., 2022a). It has a Cronbach's alpha of 0.86.

2.4. Statistical analysis

First, descriptive analyses were conducted for each dependent variable (receptive vocabulary size and vocabulary depth) and independent variables (interference control, response inhibition, VWM, NVWM, and switching), in both the TD and DLD groups. As all variables are numerical, the mean and standard deviation within each group (TD and DLD) were calculated. In addition, differences between the DLD and TD groups were explored for all variables using independent samples *t*-tests. Effect sizes (*d*) were calculated for each comparison. Following this, a Shapiro-Wilk test was used to determine whether receptive vocabulary size and vocabulary depth (the corresponding test's raw score) were normally distributed within DLD and TD groups. Because the distribution was not normal in receptive vocabulary size for TD ($W = 0.96$; $p = .004$) and DLD ($W = 0.97$; $p = .002$), and in vocabulary depth for TD group ($W = 0.97$; $p = .003$), a Spearman correlation coefficient was used to analyze the relationships among all variables within both groups.

To answer the first research question, which looked at the relationship between EF subcomponents and receptive vocabulary size in TD and DLD children, five simple regression models were estimated for each group (see Supplementary Material S1). Each EF subcomponent was included as a predictor (raw score on the corresponding task) and receptive vocabulary size as an outcome (raw score on the PPVT-III). Then, we ran two multiple regression analyses for each group separately (TD and DLD). We first ran baseline regression models with only the covariates age (in months) and non-verbal intelligence (raw score on Raven's test) as independent variables and receptive vocabulary as dependent variable (see Supplementary Material S2). In step two, we also included the EF subcomponents associated ($p < .05$) with receptive vocabulary size in the aforementioned simple regression models as predictors. We determined whether EFs explained additional variance in the model by looking at the change in variance explained (ΔR^2). Starting the analyses with simple regressions allowed us to reduce the number of predictors we introduced into the multiple regression models, increasing statistical power and reducing the possibility of collinearity.

To answer the second research question concerning the relationship between EF subcomponents and vocabulary depth in children with TD and DLD, five simple regression models were estimated for each group (TD and DLD) (see Supplementary Material S3). Each EF subcomponent was included as a predictor (raw score on the corresponding task) and vocabulary depth as an outcome (raw score on vocabulary task WISC 5). Then, we ran two multiple regression analyses for each group separately (TD and DLD). We first ran baseline regression models with only the covariates age (in months) and non-verbal intelligence (raw score on Raven's test) as independent variables and vocabulary depth as dependent variable (see Supplementary Material S4). In step two, we also included the EF subcomponents associated ($p < .05$) with vocabulary depth in the aforementioned simple regression models as predictors. We determined whether EFs explained additional variance in the model by looking at the change in variance explained (ΔR^2).

Considering the potential overlap between the different EF tasks (Friedman & Miyake, 2017), the multicollinearity was diagnosed for each multiple regression model (step 2) in the TD and DLD groups. The correlation between EF tasks was less than 0.5, and the variance inflation factor (VIF) was less than 2 for both groups. These values are indicators that none of our predictors have problematic multicollinearity (Vatcheva & Lee, 2016).

Moreover, for each of the multiple regression models run (step 2), compliance with the assumptions regarding the residuals was confirmed. Normality was assessed using the Shapiro-Wilk test, independence was evaluated with the Durbin-Watson test, and homoscedasticity was examined using the Breusch-Pagan test. All models fulfilled these three assumptions, as evidenced by non-significant *p*-values. Furthermore, the global test with 4 degrees of freedom (gvlma package) confirmed that the linear function adequately captures the estimated relationship between the variables (see Supplementary material 5 and 6). An analysis of the influence of outliers was carried out using Cook's distance and HAT methods, according to which there are no influential outliers.

Finally, to answer the third research question, related to the empirical testing of a model of the EF-vocabulary size and depth relationship in the TD and DLD groups, a path analysis was conducted using the correlation matrix. The proposed model was developed following a theoretical approach to define the direction of the relationships and an empirical approach by including only the EFs significantly associated with the vocabulary variables in the previous regression analyses. The model included the EFs as independent variables, receptive vocabulary size as a mediating variable, and vocabulary depth as a dependent variable. In addition, the model included age (in months) and non-verbal intelligence (raw score on Raven's test) as control variables. Direct, indirect, and total effects were calculated. Several fit indices were used to test the overall fit of the proposed model: Chi squared (χ^2), Comparative Fit Index (CFI), Tucker-Lewis Index (TLI), Root Mean Square Error of Approximation (RMSEA), and Standardized Root Mean Square Residual (SRMR). In addition, multivariate normality was assessed using the Mardia MVN test.

For all statistical analyses, a *p*-value < 0.05 was considered significant. The analyses were performed using R statistical software (R Core Team, 2015).

3. Results

Descriptive analyses indicate that the DLD group exhibits lower average performance than the TD group across the two vocabulary measures (receptive vocabulary size and vocabulary depth), and all tested EFs (see Table 2). When the means of both groups (TD and DLD) are compared using independent samples *t*-tests, significant differences are observed in both vocabulary knowledge measures (receptive vocabulary size and vocabulary depth) and all EFs, except switching. In the switching task, there are no significant differences between the TD and DLD groups ($t = 1.93$; $p = .06$; $d = 0.27$) (see Table 2).

3.1. Correlations among EFs, receptive vocabulary size and vocabulary depth in TD and DLD groups

Table 3 shows the correlation coefficients (Spearman's Rho) between receptive vocabulary size, vocabulary depths, and executive functions test score (interference control, response inhibition, VWM, NVWM, and switching) in the TD group. A significant correlation was found between receptive vocabulary size and response inhibition. A significant correlation was also found between vocabulary depth and tests of interference control, response inhibition, and VWM. In addition, receptive vocabulary size correlated significantly with vocabulary depths. Finally, as for the correlations between the EFs, interference control correlated significantly with response inhibition, VWM correlated significantly with interference control, response inhibition, and NVWM, and NVWM correlated

Table 2

Mean (and standard deviation) for the scores of the vocabulary (receptive vocabulary size and vocabulary depth) and executive functions tasks in TD and DLD groups. Independent samples *t*-tests for group differences are also presented.

Variables	Group		T-test		
	TD	DLD	<i>t</i> value	<i>p</i> value	Cohen's <i>D</i>
Vocabulary knowledge					
Receptive vocabulary size ^a	84.78 (13.27)	68.93 (15.64)	7.78	< 0.001	1.09
Vocabulary depth ^b	13.43 (3.96)	9.11 (3.56)	8.20	< 0.001	1.15
EFs					
Interference control ^c	21.64 (4.70)	18.85 (5.58)	3.85	< 0.001	0.54
Response inhibition ^d	70.12 (20.14)	59.09 (25.77)	3.39	< 0.001	0.48
VWM ^e	5.58 (1.67)	4.03 (2.35)	5.39	< 0.001	0.76
NVWM ^f	3.07 (1.52)	2.28 (1.69)	3.52	< 0.001	0.49
Switching ^g	11.88 (3.46)	10.86 (4.11)	1.93	.056	0.27

Note. TD: Typical development; DLD: Developmental language disorder; VWM: Verbal working memory; NVWM: Nonverbal working memory. Statistically significant difference in bold.

^a Raw score on the Peabody Picture Vocabulary test (PPVT-III). Higher scores indicate better performance for receptive vocabulary size.

^b Raw score on the Vocabulary task (WISC 5). Higher scores indicate better performance for vocabulary depth.

^c Raw score on the Arrows task. Higher scores indicate better performance for interference control.

^d Raw score on the Flies task. Higher scores indicate better performance for response inhibition.

^e Raw score on the Recall of Digits Backward subtest. Higher scores indicate better performance for verbal working memory.

^f Raw score on the Visual Farm task. Higher scores indicate better performance for no verbal working memory.

^g Raw score on the Triads task. Higher scores indicate better performance for switching.

Table 3

Correlation coefficients (Spearman's Rho) between vocabulary test results (receptive vocabulary size and vocabulary depths) and executive functions tasks in the TD group ($N = 99$).

Variable	1	2	3	4	5	6
1 Receptive vocabulary size						
2 Vocabulary depth	.44***					
3 Interference control	.18	.22*				
4 Response inhibition	.23*	.27**	.40***			
5 VWM	.07	.21*	.23*	.24*		
6 NVWM	.00	.11	.14	.24*	.48***	
7 Switching	.04	.03	−0.02	−0.07	−0.23*	−0.10

* $p < .05$;

** $p < .01$;

*** $p < .001$

VWM: Verbal working memory; NVWM: Nonverbal working memory.

significantly with response inhibition. All these correlations are positive. In contrast, switching was significantly and negatively correlated with VWM.

Table 4 show the correlation coefficients (Spearman's Rho) between receptive vocabulary size, vocabulary depth, and executive functions test scores (interference control, response inhibition, VWM, NVWM, and switching) in the DLD group. A significant correlation was found between receptive vocabulary size and tests of interference control, VWM and switching. A significant correlation was also found between vocabulary depth and tests of interference control, response inhibition, and VWM. In addition, receptive vocabulary size correlated significantly with vocabulary depth. Finally, as for the correlations between the EFs, interference control correlated significantly with response inhibition, VWM and NVWM, VWM correlated significantly with interference control, response inhibition, and NVWM, NVWM correlated significantly with interference control, response inhibition, VWM and switching. All these correlations are positive.

3.2. Research question 1: relationship between EF subcomponents and receptive vocabulary size in TD and DLD children

The baseline model for the TD and DLD groups, including the covariates (age and non-verbal intelligence) as independent variables and receptive vocabulary size as dependent variable, is not presented here (see Supplementary Material S2).

In the TD group, including the EF subcomponents as predictors to the baseline model with receptive vocabulary size as the dependent variable did not lead to a significant increase in the variance explained ($\Delta F(2, 94) = 0.48, p = .620, \Delta R^2 = 0.0$). None of the EF subcomponents was significantly associated with receptive vocabulary size. All variables together (i.e., age, non-verbal intelligence, interference control, and response inhibition) explained 22% of the variance in receptive vocabulary size in the TD group (see Table 5). In the DLD group, including the EFs as predictors to the baseline model with receptive vocabulary size as the dependent variable lead to a significant increase in the variance explained ($\Delta F(4, 98) = 3.65, p = .008, \Delta R^2 = 0.8$). VWM and switching were significantly associated with receptive vocabulary size in this group. All variables together (i.e., age, non-verbal intelligence, interference control, VWM, NVWM and switching) explained 23% of the variance in receptive vocabulary size in the DLD group (see Table 5).

3.3. Research question 2: relationship between EF subcomponents and vocabulary depth in TD and DLD children

The baseline model for the TD and DLD groups, including the covariates (age and non-verbal intelligence) as independent variables and vocabulary depth as dependent variable, is not presented here (see Supplementary Material S4).

In the TD group, including the EF subcomponents as predictors to the baseline model with vocabulary depth as the dependent

Table 4

Correlation coefficients (Spearman's Rho) between vocabulary test results (receptive vocabulary size and vocabulary depth) and executive functions tasks in the DLD group ($N = 105$).

Variable	1	2	3	4	5	6
1 Receptive vocabulary size						
2 Vocabulary depth	.54***					
3 Interference control	.20*	.23**				
4 Response inhibition	.06	.16*	.29**			
5 VWM	.35***	.32**	.32**	.30**		
6 NVWM	.26**	.03	.20*	.27**	.35***	
7 Switching	.27**	.19	.13	.10	.15	.23*

* $p < .05$;

** $p < .01$;

*** $p < .001$

VWM: Verbal working memory; NVWM: Nonverbal working memory.

Table 5

Multiple linear regression for receptive vocabulary size (dependent variable), non-verbal intelligence and age (covariates), and executive functions (independent variables) in TD and DLD children.

TD group (n = 99)					
Predictors	B	CI (95%)	β	p	Adj. R2
Step 2					.220
Age (months)	.619	.240 – 0.998	.343	.002	
Non-verbal intelligence	.631	.033 – 1.230	.209	.039	
Interference control	.269	–0.294 - 0.831	.095	.346	
Response inhibition	–0.004	–0.149 - 0.141	–0.006	.957	
DLD group (n = 105)					
Predictors	B	CI (95%)	β	p	Adj. R2
Step 2					.226
Age (months)	.471	.041 - 0.900	.210	.032	
Non-verbal intelligence	–0.242	–1.150 - 0.666	–0.056	.598	
Interference control	.358	–0.171 - 0.887	.128	.183	
VWM	1.708	.355 – 3.060	.257	.014	
NVWM	.212	–1.576 – 1.999	.023	.815	
Switching	.744	.025– 1.463	.196	.043	

Note. Step 1, the regression model only containing the covariates can be found in Supplementary Material 2. Statistically significant associations in bold. Only EF subcomponents significantly associated with receptive vocabulary size and vocabulary depth ($p < .05$) in previously conducted simple regressions were included. VWM: Verbal working memory; NVWM: Nonverbal working memory. All coefficients are presented as unstandardized (B) and standardized (β).

variable did not lead to a significant increase in the variance explained ($\Delta F(3, 93) = 0.95, p = .420, \Delta R^2 = 0.0$). None of the EF subcomponents was significantly associated with vocabulary depth in the TD group. All variables together (i.e., age, non-verbal intelligence, interference control, response inhibition, and VWM) explained 10% of the variance in vocabulary depth in the TD group (see Table 6). In the DLD group, including the EFs as predictors to the baseline model with vocabulary depth as the dependent variable did not lead to a significant increase in the variance explained ($\Delta F(3, 99) = 1.22, p = .310, \Delta R^2 = 0.1$). None of the EF subcomponents was significantly associated with vocabulary depth in the DLD group. All variables together (i.e., age, non-verbal intelligence, interference control, VWM, and switching) explained 28% of the variance in vocabulary depth in the DLD group (see Table 6).

3.4. Research question 3: model of the EF-receptive vocabulary size-vocabulary depth relationship in the TD and DLD groups

In the TD group, no path analysis was run as the previously performed multiple regressions found no relationship between EFs and vocabulary knowledge (receptive vocabulary size and vocabulary depth). Fig. 1 shows the proposed model in the DLD group. This model presents excellent fit indicators ($\chi^2(3) = 0.097, p = .992$; CFI = 1; TLI = 1.125; RMSEA < 0.001; SRMR = 0.005), suggesting that it is parsimonious and accurately reproduces the relationships between the studied variables. This model was identified as having 3

Table 6

Multiple linear regression for vocabulary depth (dependent variable), non-verbal intelligence and age (covariates), and executive functions (independent variables) in TD and DLD children.

TD group (n = 99)					
Predictors	B	CI (95%)	β	p	Adj. R2
Step 2					
Age (months)	.105	–0.016 - 0.227	.195	.089	
Non-verbal intelligence	.069	–0.138 - 0.276	.077	.510	
Interference control	.064	–0.116 - 0.245	.076	.481	.101
Response inhibition	.019	–0.027 - 0.066	.098	.415	
VWM	.236	–0.274 - 0.747	.100	.360	
DLD group (n = 105)					
Predictors	B	CI (95%)	β	p	Adj. R2
Step 2					
Age (months)	.213	.119 - 0.310	.418	<0.001	
Non-verbal intelligence	.054	–0.140 - 0.25	.054	.584	
Interference control	.046	–0.070 - 0.160	.073	.431	.276
VWM	.218	–0.073 - 0.510	.144	.140	
Switching	.044	–0.113 - 0.200	.050	.579	

Note. Step 1, the regression model only containing the covariates can be found in Supplementary Material 4. Statistically significant associations in bold. Only EF subcomponents significantly associated with receptive vocabulary size and vocabulary depth ($p < .05$) in previously conducted simple regressions were included. VWM: Verbal working memory. All coefficients are presented as unstandardized (B) and standardized (β).

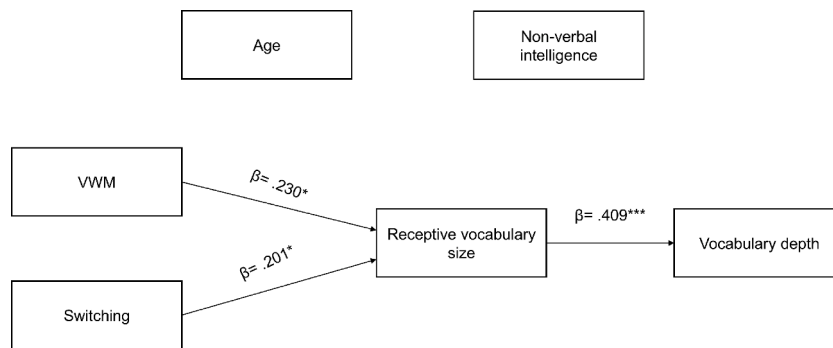


Fig. 1. Path analysis for direct and indirect effects between executive functions, receptive vocabulary size and vocabulary depth in DLD group. The boxes at the top of the figure represent the control variables. All coefficients are standardized. * $p < .05$, ** $p < .01$, *** $p < .001$.

degrees of freedom available. In addition, the results obtained indicate multivariate normality (Mardia Skewness=140.92, p -value=0.09; Mardia Kurtosis= -0.14, p -value=0.89).

The path analysis shows that VWM has a positive and significant direct effect on receptive vocabulary size ($\beta=0.23$; $p = .02$). Similarly, switching has a direct effect on receptive vocabulary size ($\beta=0.20$; $p = .03$). Together, VWM, switching, and the covariates age and non-verbal intelligence explain 21% of the variance in receptive vocabulary size ($R^2 = 0.207$). In addition, receptive vocabulary size has a direct effect on vocabulary depth ($\beta = 0.41$, $p < .001$). Together, receptive vocabulary size plus the covariates age and non-verbal intelligence explain 41% of vocabulary depth ($R^2 = 0.407$) (see Fig. 1). Finally, the path analysis shows that receptive vocabulary size mediates between EFs and vocabulary depth. In this sense, the indirect effect of VWM on vocabulary depth is positive and significant ($\beta = 0.09$, $p < .001$), as is the indirect effect of switching on vocabulary depth ($\beta = 0.08$, $p < .001$).

4. Discussion

The current study found that children with DLD understand fewer words (receptive vocabulary size) and the definitions they provide are less precise (vocabulary depth) than same age TD children. This supports previous studies on English-speaking children with DLD (Marinellie & Johnson, 2002; McGregor et al., 2013; Rice & Hoffman, 2015) and on children with DLD learning Spanish (Acosta et al., 2019; Coloma et al., 2020). Additionally, in support of the wider literature (Henry et al., 2012; Pauls & Archibald, 2016; Vugs et al., 2014), the DLD group had poorer EF performance than the TD children in all components other than the switching task. Given that the majority of EF tasks were presented non-verbally, except for verbal working memory, these difficulties are unlikely to come from the language demands of the tasks themselves (Kapa et al., 2017).

Supporting previous research (e.g., Nippold, 2007; Stahl & Nagy, 2006), we found that for both groups, performance in receptive vocabulary size and vocabulary depth were correlated with a medium-large effect size (TD: $Rho = 0.44$, $p < .001$; DLD: $Rho = 0.55$, $p < .001$). This supports the idea that receptive vocabulary size and vocabulary depth are two distinct but associated aspects of vocabulary and both should be examined when exploring vocabulary knowledge in children.

We investigated the role of EF on vocabulary knowledge by asking three main questions:

1. Is there a relationship between EF subcomponents and receptive vocabulary size in children with TD and DLD, after accounting for participants' age and non-verbal intelligence? If so, which EF subcomponents best explain the variability of receptive vocabulary size in each group, TD and DLD?

First, as hypothesized, our results confirmed a relationship between EF subcomponents and receptive vocabulary size in the DLD group after controlling for participants' age, and non-verbal intelligence. These findings are consistent with previous research on school-aged children with DLD (Jackson et al., 2016; Vugs et al., 2017). As hypothesized, the results of the current study also indicate that EFs do not significantly explain the extent of variance in receptive vocabulary size in TD school-aged children. This finding also aligns with previous research in TD children of the same age range (Kaushanskaya et al., 2017). Kuhl (2004), argued that as language develops, children show a decreasing reliance on other cognitive functions such as EFs when carrying out linguistic related tasks. Therefore, as other authors have suggested, it is possible that the relationship between EFs and receptive vocabulary size in children with TD is stronger in the early years of development (Everaert et al., 2023; White et al., 2017). This relationship continues at later ages in children with DLD, as a possible indicator of the greater cognitive effort required for language processing (Blom & Boerma, 2019; Everaert et al., 2023). It is possible that in TD school-aged children, access to the lexicon and processing of vocabulary meanings has become automatic and less reliant on EF. This automatization appears delayed in children with DLD who carry on relying on EFs for these tasks.

Second, our results show that the EFs VWM and switching significantly explain receptive vocabulary size in children with DLD, over and above what is explained by participants' age, and non-verbal intelligence. The findings are consistent with our original hypotheses in that previous studies have found a relationship between VWM and receptive vocabulary size in preschool children with DLD (Horohov & Oetting, 2004; Vugs et al., 2016), a relationship that continues through to school age in this group (Jackson

et al., 2016; Vugs et al. 2017). These results also support the theoretical framework we adopted that argues for a relationship between working memory and vocabulary acquisition. Several authors have proposed that VWM plays a crucial role in vocabulary learning, particularly during the early stages of language acquisition (Gathercole, 2006; Archibald, 2017). VWM allows for the temporary retention of new phonological forms in the phonological loop during learning, which is essential for creating stable word representations through repeated exposures (Gathercole, 2006). In the initial stages of acquisition, new word learning in young children depends on the brief encoding of phonological forms in VWM. However, as children grow older and their lexicons expand, vocabulary processing is facilitated by automatic access to lexical representations of similar words and less on real-time processing in VWM (Archibald, 2017). In contrast, children with DLD continue to need this support from VWM for vocabulary learning during the school years.

Regarding the role of cognitive flexibility, contrary to our hypotheses, the results show that switching was also significantly associated with receptive vocabulary size, but only in children with DLD. These results do not align with those of Kalliontzi et al. (2022) in preschool children with DLD. The age range of the children may underlie the differences between the two studies. In our DLD sample, children were significantly older than in Kalliontzi et al. (2022). Therefore, considering that cognitive flexibility develops at a slower rate than VWM (Diamond, 2013), it is possible that finding a relationship between switching and vocabulary knowledge requires older samples than the one included in the Kalliontzi et al. (2022) study. Interestingly, switching was the only EF in our research where children with DLD performed similarly to TDs. From this finding, we suggest switching acts as a compensatory mechanism in conjunction with VWM. In line with our findings, some authors have already proposed that children with DLD use compensatory mechanisms in language tasks (see Blom & Boerma, 2019; Kalliontzi et al., 2022). Based on these findings, we encourage future research to study further the role of the most preserved EFs within the DLD group as compensation in the language development of these children. As indicated in the introduction, very few studies have explored the role of cognitive flexibility in vocabulary development. For example, Hill and Wagovich (2020) found that cognitive flexibility supports word learning in school-aged children, particularly in contexts that require switching between multiple simultaneous tasks. Notably, these authors also identified a compensatory mechanism whereby children with weaker language skills but stronger cognitive flexibility showed greater word learning gains, while those children with stronger language skills relied less on cognitive flexibility. Future studies should further explore the role of cognitive flexibility in the vocabulary knowledge of children with DLD.

Interestingly, we did not find a relationship between the two inhibitory control measures (interference control and response inhibition) and vocabulary size in our school-aged children with TD and DLD. These results contradict the notion that inhibitory control, with its two subcomponents, is a relevant element for vocabulary development (e.g., Gandolfi & Viterbori, 2020). The difference between our results and others could be explained by the idea that inhibitory control plays a more important role at younger ages than the children in our sample (e.g., Spaulding, 2010). A study by Kapa and Erikson (2020) found that inhibitory control was associated with word learning in children with TD and DLD; but only in preschool children. Previous studies with school-aged children have found results similar to ours: in TD (e.g., Kaushanskaya et al., 2017) and DLD children (e.g., Larson et al., 2020). It is also important to note that our study included several EFs in the same analysis model, thereby controlling for potential covariation effects between EFs. If the relationship between inhibitory control and vocabulary knowledge is assessed without including other EFs, as other studies have done, it is not possible to determine the role of the other unmeasured EFs.

2. Is there a relationship between EF subcomponents and vocabulary depth in children with TD and DLD, after accounting for participants' age and non-verbal intelligence? If so, which EF subcomponents best explain the variability of vocabulary depth in each group, TD and DLD.

As we hypothesized, our results indicate that no significant relationship exists between EF subcomponents and vocabulary depth in children with DLD when controlling for participants' age, and non-verbal intelligence. This is consistent with previous studies on school-aged children with DLD (Dosi & Gavriilidou, 2020; Dosi, 2021). However, contrary to our hypotheses, our findings show that there are also no significant relationships between EF subcomponents and vocabulary depth in the TD children either. While previous studies report a significant relationship (e.g., Dosi and Gavriilidou, 2020), it has also been reported that this relationship takes some time to emerge. Dosi (2021) reports a significant relationship between VWM and the ability to make definitions in TD children by 8:5 years, but not in younger groups. Considering that our sample of TD children had an average age of 7:3 years old, it is possible that the EF-vocabulary depth relationship is still to emerge. In addition, previous studies have shown that vocabulary depth depends on other linguistic factors, such as receptive vocabulary size and semantic and syntactic knowledge (Gutierrez-Cleffen & DeCurtis, 1999). The absence of a relationship between EFs and vocabulary depth may indicate that in children with TD and DLD, other linguistic skills may play a more relevant role than EFs.

The next question asked:

3. If there is a relationship between EF, vocabulary size and depth in children with TD and DLD, does vocabulary size play a mediating role in the relationship between EF and vocabulary depth?

As previously indicated, in the TD group no path analysis was carried out, as we found no relationship between EFs and receptive vocabulary size or vocabulary depth. These findings suggest that the cognitive mechanism involved in vocabulary processing for TD school children is functioning largely independent of EFs.

In the DLD group, the path analysis shows a direct and positive association between the EFs VWM and switching with receptive vocabulary size. The effect of both EFs on vocabulary size has a similar magnitude ($\beta = 0.23$ for VWM and $\beta = 0.20$ for switching). These findings suggest that VWM is still employed during the acquisition of new words in school-aged children with DLD. As VWM is the most delayed EF in children with DLD it is possible that cognitive flexibility is acting as a compensatory mechanism, as we described for the previous research question. Our data also show that EFs, together with the covariates of age and nonverbal

intelligence, explain 21% of the variance in receptive vocabulary size. These findings suggest that while EFs play a relevant role in vocabulary development, other variables not measured in this study could also account for the variability in receptive vocabulary size. Other studies have identified that variables such as metalinguistic awareness (e.g., Altman et al., 2018) and auditory speech perception (Erdener & Burnham, 2018) can also influence children's vocabulary size.

As we hypothesized, the path analysis also shows an indirect and positive relationship between VWM and switching with vocabulary depth and this is mediated by receptive vocabulary size. According to these findings, the effect of EFs on vocabulary depth is linked to the receptive vocabulary size of the children. A significant percentage (41%) of the variability in vocabulary depth is explained by receptive vocabulary size and the covariates of age and nonverbal intelligence. However, as mentioned earlier, other variables not measured in this study could also have influenced the children's depth of lexical knowledge.

The EF-vocabulary size and depth model proposed in this study is exploratory, especially given that, as a whole, these relationships have not been previously studied in children with DLD. Therefore, other explanatory paths between the variables could also exist. In addition, in the wider literature there is some debate about the direction of the relationships between EF and vocabulary (e.g., Schmitt et al., 2019; Slot & Von Suchodoletz, 2018), as well as, between both aspects of vocabulary—size and depth (Li & Kirby, 2015). In this sense, other models involving variables not included here or relationships in different directions than those proposed in this study, could also fit the data well. However, a cross-sectional design, such as in our study, does not allow us to determine directionality and therefore the path analysis should be interpreted cautiously. Future research could investigate these relationships and compare them with other models using longitudinal or experimental designs to establish causal relationships between variables (Shadish et al., 2002).

4.1. Limitations

This study has some limitations. First, there is an ongoing discussion in the literature concerning the overlap between the EFs and how tasks may tap into a number of EFs at the same time (see Friedman and Miyake, 2017). One suggestion to clarify this is to use several tasks for each measured EF and perform a factor analysis. However, this was not feasible in our study because of testing overload, as the children were already doing several different tasks. Another possibility is to include all tasks in a single latent variable, as other studies have done (e.g., Blom and Boerma, 2019; Everaert et al., 2023). However, this would not have allowed us to identify the specific EF subcomponents associated with vocabulary knowledge, which was our main aim. In order to try and isolate EFs, we selected classical tasks (e.g., backward digits for VWM) or adaptations of classical tasks (the Yellow-Red battery) with high-reliability indices to measure each EF subcomponent.

A second potential limitation of the study concerns the measurement of vocabulary depth. Some authors suggest using high-frequency (i.e. familiar) words to measure vocabulary depth to better isolate this aspect from vocabulary size. Using familiar words would ensure a more accurate assessment of children's depth of knowledge of the acquired lexicon (e.g., Zhang & Koda, 2017). However, achieving completely independent measures is challenging. Our study did not control word frequency on the vocabulary depth task, as we prioritized using a standardized test appropriate to our context. Future research could address this issue by employing tests or methodologies that account for words familiarity.

A third potential limitation of the study is the requirement for verbal responses in the vocabulary depth task. As some authors have suggested, using tests that demand verbal responses may underestimate performance in children with language disorders (e.g., Drevon et al., 2017). For this reason, the measure of vocabulary depth in children with DLD may be influenced by their existing language production problems. As previously indicated, we used the vocabulary depth task as it is standardized for our setting but future studies should consider using nonverbal response formats to avoid this issue. Another potential limitation of the study is that our sample is limited to one age range (6–8 years) and only families from low SES. More studies are needed in other age ranges and SES levels before generalizing our results.

4.2. Theoretical and clinical implications

From a theoretical point of view, this study contributes to the ongoing debate about the cognitive mechanisms underlying vocabulary development, both in children with TD and in DLD. The results show that not all EF subcomponents are equally related to vocabulary skills in DLD children, an issue that may be overlooked when EFs are assessed as a single EF construct. Furthermore, the findings show that EFs are not equally related to different aspects of vocabulary knowledge in children with DLD. Based on these findings, we argue that future research should consider the multidimensional nature of vocabulary development (Koizumi & In'nami, 2020). Our study supports the idea that the relationship between EF and vocabulary acquisition in school-aged children with DLD is complex. EF abilities continue to be active in a similar way to younger TD children and EFs are used as compensatory mechanisms to make up for language processing difficulties.

From a clinical perspective, this study has important implications for Speech and Language Pathologists (SLPs) and others involved in supporting language development in children with DLD, such as teachers. Specifically, if the goal is to enhance vocabulary knowledge in children with DLD, it is crucial also to consider EFs in detail. Based on the current study's findings, SLPs might want to prioritize the EFs of VWM and cognitive flexibility during language therapy. Planning interventions that target both language structures and specific EFs could strengthen the relationship between these areas. As vocabulary develops it is possible it will reciprocally support the EFs and this relationship might work in the opposite direction. In this regard, recent research has shown that interventions targeting VWM can positively impact vocabulary knowledge (e.g., Acosta et al., 2019).

5. Conclusions

This study provides new evidence on the complex relationship between EFs and vocabulary knowledge in Spanish-speaking school aged children with and without DLD. In children with DLD, our results indicate that VWM and switching are directly associated with receptive vocabulary size and are indirectly associated with vocabulary depth. All the relationships found are positive, i.e., as vocabulary size increases, so does depth. In TD children, no EF subcomponent was significantly associated with either aspect of vocabulary, suggesting this group is using language more automatically and efficiently than the DLD children of the same ages. These results underline the importance of considering EFs as factors underlying vocabulary knowledge in DLD.

Research data

The data and code necessary to reproduce the analyses presented here are available from the corresponding author on reasonable request, as are the materials necessary to attempt to replicate the findings. The research data and scripts are not in a data repository due to the disposition of the ethics committee.

Funding

This research was supported by the Agencia Nacional de Investigación y Desarrollo (ANID, Government of Chile) under Grant PIE CIE 160007 and individual grant 242230153. Funding from CEDETI UC is also gratefully acknowledged.

CRediT authorship contribution statement

Felipe Torres-Morales: Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Gary Morgan:** Writing – review & editing, Writing – original draft. **Ricardo Rosas:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the school principals for facilitating access to the participants. We appreciate the time and effort of participants and their families.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jcomdis.2025.106498](https://doi.org/10.1016/j.jcomdis.2025.106498).

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